in the famous periodic table which is pinned to the walls of so many classrooms and which summarizes the properties of the elements of which the world is made – not only on Earth but all over the universe in all galaxies. Why these specific elements? What explains the periodic structure of the table? Why does each element have certain properties and not others? Why, for instance, do some elements combine easily, whereas others do not? What is the secret of the curious structure of Mendeleev's table?

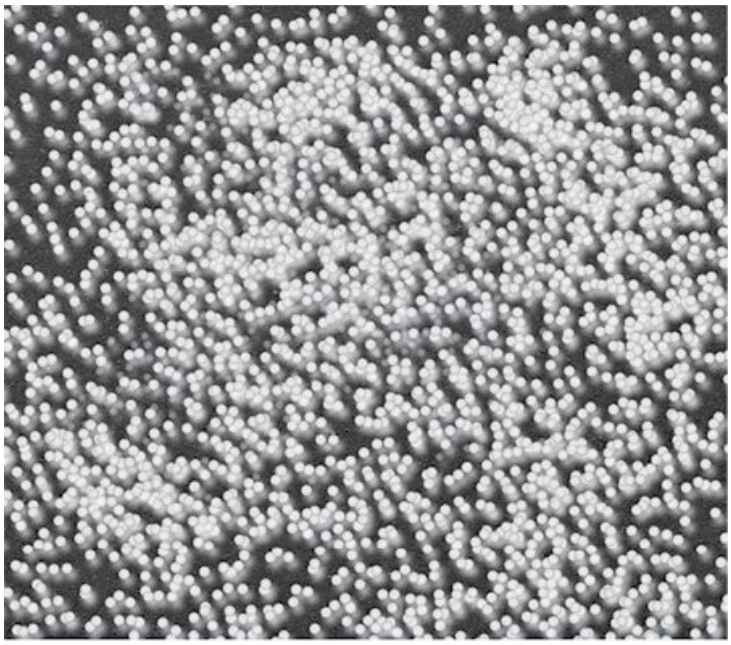


Figure 4.5 Light is a wave on a field, but it has also a granular structure.

Well, take the equation of quantum mechanics that determines the form of the orbitals of an electron. This equation has a certain number of solutions, and these solutions correspond exactly to hydrogen, helium, oxygen … and the other elements! Mendeleev's periodic table is structured exactly like the set of these solutions. The properties of the elements, with everything else, follows from the solution of this equation. Quantum mechanics deciphers perfectly the secret of the structure of the periodic table of elements.

Pythagoras and Plato's ancient dream is realized: to describe all of the world's substances with a single formula. The infinite complexity of chemistry, captured by the solutions of a single equation! And this is just one of the applications of quantum mechanics.

Fields and particles are the same thing

Shortly after completing the general formulation of quantum mechanics, Dirac realizes that the theory can be directly applied to fields such as electromagnetic ones, and can be made consistent with special relativity.